NullState: a reference-anchored framework for detecting, diagnosing, and quantifying the off-target compartment of human neural organoids

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Abstract

Human neural organoids reproducibly generate cell types that lie outside the developmental trajectories they are intended to model, but the field lacks a principled, atlas-anchored way to *quantify* this off-target compartment and ask whether it carries reproducible structure. We present **NullState**, a reference-based pipeline that maps the 1,767,346-cell Human Neural Organoid Atlas (HNOCA) onto a harmonized 2,615,898-cell first-trimester human fetal reference (HDBCA + a fetal-body atlas) by variational reference surgery, assigns each organoid cell a pair of calibrated mapping scores (off-manifold distance and label entropy), and classifies it into a six-state off-target taxonomy. Across the atlas, 38.4% of cells fall into a large, *largely neural* ambiguous-novel (off-manifold) compartment; quantifying and explaining it is the work of Aim 2. Four orthogonal diagnostics show it is not a projection artifact but a *structured mixture* of a reference-coverage gap and a pervasive in-vitro stress program: off-manifold cells lose neural-progenitor identity while gaining hypoxia and unfolded-protein-response signatures, their burden rises monotonically with maturation and metabolic load, varies ∼240-fold across differentiation protocols (0.4% to 96.8%), and concentrates into coherent transcriptomic clusters; projecting these cells onto an *independent* prenatal cortex atlas confirms a genuine in-vitro-divergence component beyond the reference gap (ambiguous-novel cells sit further from that atlas than clean cells, Cliff's $\delta = 0.30$). Asking next whether lineage can be separated from the organoid-versus-primary shift, we report a reproducible *negative* result: a shared-latent contrastive model cannot. An origin-decoding adversary recovers organoid-versus-primary identity from the putatively dish-stripped latent at ≥ 0.998 balanced accuracy, robustly across latent-capacity and batch-conditioning interventions. Because the two sources occupy near-disjoint expression manifolds, this shift—biological in-vitro state and technical platform differences combined—is pervasive and encoder-visible, not a separable nuisance subspace; we therefore state the result methodologically (shared-latent methods cannot yield a source-invariant lineage axis here) and read the specifically in-vitro biology from the stress diagnostics rather than from the adversary. Finally, on the *non-neural* contaminant populations—where, unlike the reference-limited neural compartment, we are statistically powered—we show the geometry method works as an instrument: pairwise distances clear a matched-N noise floor (14–105×) and their *rank order* recovers the primary lineage geometry (distance-matrix $r = 0.90$). A direct test, however, finds the per-lineage source offsets are not constant (mutual cosine 0.15), so the individual distances are confounded rather than clean lineage metrics—a validated ordering, not yet a clean geometry. NullState thus provides a reference-anchored, *comparative* off-target measurement—to be read against the reference's own domain, not as an absolute fidelity score—and an identifiability boundary for when latent calibration without anchoring is attainable at all.

1 Introduction

Human pluripotent-stem-cell-derived neural organoids have become a central model of human brain development [1, 2], capturing cortical cytoarchitecture, diverse neuronal and glial identities, and functional network activity [3–5]. Yet a recurring concern limits their interpretation: organoids also produce cells that do not correspond to any *bona fide* in-vivo neural state. These include non-neural lineages arising from incomplete germ-layer restriction and aberrant states driven by the culture environment itself. A landmark observation is that metabolic and endoplasmic-reticulum (ER) stress in organoids impairs the specification of correct molecular subtypes [6], and cross-protocol comparisons against fetal tissue have repeatedly surfaced protocol-dependent, off-target differentiation routes [7, 8]. The recent Human Neural Organoid Atlas (HNOCA)—an integrated transcriptomic atlas of >1.7 million organoid cells spanning dozens of protocols [9]—makes it possible, for the first time, to study this off-target compartment at scale and against a common coordinate system.

What has been missing is a *principled, reference-anchored* way to (i) decide which organoid cells are off-target, (ii) explain *why* they are off-target, and (iii) ask whether the off-target compartment is merely noise or carries reproducible biological structure. Reference-based mapping provides the natural substrate: by projecting organoid cells onto a comprehensive fetal atlas, cells that map confidently to a fetal identity can be separated from those that do not. Two recently completed human references make such anchoring feasible—a comprehensive first-trimester human brain cell atlas (HDBCA) [10] and a human fetal-body gene-expression atlas [11]—together spanning neural and non-neural developmental origins.

Here we introduce **NullState**, a pipeline that operationalizes this idea (Fig. 1). We harmonize the two fetal atlases into a single 2,615,898-cell reference, map the 1,767,346-cell HNOCA query onto it by variational reference surgery [12–14], and assign every organoid cell two calibrated mapping scores—an off-manifold reconstruction distance and a label-assignment entropy—that jointly define a six-state off-target taxonomy. We organize the analysis around three questions: **(Aim 1)** is the off-target measurement well-posed and adequately powered? **(Aim 2)** what *is* the dominant, largely neural off-manifold compartment, and can lineage be separated from the in-vitro state so that those cells can be placed on a "dish-stripped" lineage axis? and **(Aim 3)** do off-target populations occupy distinct, reproducible geometry? The division of labor between the last two is deliberate and worth stating up front. Aim 2 quantifies and explains the neural off-manifold compartment, the bulk of the off-target signal but one that the present first-trimester reference cannot yet resolve into clean lineage geometry; Aim 3 then turns to the *non-neural* contaminant populations (skeletal and cardiac muscle, fibroblast, adrenal-cortical cells), the compartment where the reference *is* complete and the cell numbers *are* sufficient, and asks whether the method recovers real lineage structure there. Our central findings are that the off-manifold compartment is a structured biological mixture rather than an artifact; that shared-latent disentanglement provably fails here for an interpretable reason, so the negative is stated methodologically and the biology read from the diagnostics; and that off-target geometry, demonstrated on the powered non-neural compartment, is recoverable and biologically coherent.

2 Results

2.1 A harmonized fetal reference supports confident organoid mapping

We assembled a fetal reference by combining the first-trimester HDBCA [10] with the non-neural compartments of a human fetal-body atlas [11], retrieving raw unique-molecular-identifier (UMI) counts from the CZ CELLxGENE Discover Census [15]. The merged reference comprises 2,615,898 cells $\times$ 61,497 genes, with 32/33 lineage origins resolved. We trained a single-cell variational model (scVI) for 400 epochs (evidence lower bound, ELBO, 695.6), refined it into a label-aware model (scANVI, ELBO 736.1), and projected the HNOCA query onto the reference by architectural surgery (scArches; query-surgery ELBO 920.2) [12–14, 16]. Surgery required two stabilizers that we found essential at atlas scale: excluding 9,229 non-UMI (full-length read-count) cells whose library sizes destabilize the negative-binomial encoder, and filtering to cells with ≥ 50 counts across highly variable genes (1,767,346 $\rightarrow$ 1,689,762; 4.4% removed).

Donor-level batch integration improved cross-batch mixing while preserving biological structure: the integration local inverse Simpson's index (iLISI) rose from 6.10 (unintegrated principal components) to 8.10 after scVI, with only a modest change in the cell-type cLISI (1.31 $\rightarrow$ 1.44) [17, 18] (Fig. 2c). We calibrated two decision thresholds on held-out reference cells—a label-entropy threshold $\tau_H = 0.96$ and an off-manifold threshold $\tau_R = 1.93$—and used them to assign each organoid cell to one of six states (Fig. 2a,b). The resulting partition is dominated by an *ambiguous-novel* (off-manifold) compartment (677,783 cells; 38.4%), followed by clean on-target (459,441; 26.0%), poorly differentiated on-target (281,217; 15.9%), ambiguous (167,838; 9.5%), true off-target (93,795; 5.3%), and quality-control-dropped cells (86,813; 4.9%).

Three feasibility gates confirmed the measurement is well-posed (Fig. 2d). The off-target taxonomy is internally consistent (Gate A, pass); an in-vitro "dish-vector" computed on clean on-target cells is highly self-consistent (Gate B, mean cosine $0.855 > 0.70$), justifying a single shared in-vitro axis; and the true off-target compartment is adequately powered for downstream geometry (Gate C, green; 93,795 cells). NullState thus yields a calibrated, adequately powered off-target measurement over the entire organoid atlas.

2.2 The off-manifold compartment is a structured mixture of reference gap and in-vitro stress

The size of the ambiguous-novel compartment (38.4%) raises an essential question: is it a mapping artifact, or does it reflect real biology? Four orthogonal diagnostics, computed on the neural subset, converge on the latter (Fig. 3).

Identity and stress programs (Fig. 3a). Relative to clean on-target neural cells, ambiguous-novel neural cells show markedly reduced neural-progenitor identity (NPC score 1.11 $\rightarrow$ 0.61) while gaining canonical stress signatures: hypoxia (0.60 $\rightarrow$ 0.96) [19] and the unfolded-protein/ER-stress response (0.12 $\rightarrow$ 0.56, a $\sim$5-fold increase). An independent, atlas-precomputed glycolysis hallmark score is correspondingly elevated (0.46 $\rightarrow$ 0.65). This is the transcriptional signature of culture stress described by Bhaduri *et al.* [6], here resolved at atlas scale and tied directly to off-manifold status.

Maturation and metabolic load (Fig. 3b). The off-manifold fraction rises monotonically with a per-cell potency/complexity proxy (number of detected genes; quintile fractions 0.13 $\rightarrow$ 0.78) and is highest in the oldest organoids (age quintile fraction 0.62; median off-manifold score 1.84 $\rightarrow$ 2.12 across age) [20]. Off-target burden therefore accrues with culture time and metabolic demand, consistent with a stress-driven mechanism rather than random misassignment.

Protocol dependence (Fig. 3c). Across 26 differentiation protocols, the ambiguous-novel fraction spans nearly the entire range (0.4% to 96.8%; standard deviation 0.28). Guided cortical protocols sit at the low end (e.g., DAPT-patterned cortico-motor [21], 0.4%; 2, 10%), whereas regionally divergent or barrier-forming protocols dominate the high end—choroid-plexus organoids [22] reach 96.8%, midbrain-dopaminergic [23] and thalamic [24] protocols 85–90%. The off-manifold signal thus partly tracks *reference coverage*: the first-trimester, cortex-rich HDBCA under-represents later and non-cortical neural states, which are then read as off-manifold.

Cluster coherence (Fig. 3d,e). Ambiguous-novel cells are not scattered. Across 30 Leiden communities in the atlas embedding [25], the off-manifold fraction is bimodal (per-cluster s.d. 0.29 about an overall mean of 0.55); 20% of cells reside in near-pure off-manifold clusters ($\geq 80\%$) and 21% in near-pure clean clusters ($\leq 20\%$). Off-manifold status is a coherent population-level property, visible as contiguous territory in the scPoli UMAP [26].

Direct cross-atlas test (Fig. 6). The protocol gradient argues for a coverage-gap contribution indirectly; we tested it directly by asking whether off-manifold neural cells map onto an *independent*, non-HDBCA reference—the prenatal human cortex of Velmeshev *et al.* [27]—reasoning that cells merely missing from HDBCA should be recoverable there, whereas genuinely divergent cells should not. The experiment exposes, and is bounded by, the same pervasiveness that defeats Aim 2: the organoid-versus-primary source gap displaces *every* organoid cell from the independent atlas, including clean on-target cells that map confidently to HDBCA (clean median off-manifold distance 4.0 versus reference-self 2.2), under both scArches projection and joint scVI integration. Absolute "mapped-on" fractions are therefore confounded, and we do not report them. Instead the clean control *quantifies* that shared displacement, and—exactly as in the Aim 3 cancellation argument—the clean-versus-ambiguous-novel contrast removes it: ambiguous-novel neural cells are significantly and consistently further from the independent cortex manifold than clean cells (area under the curve 0.65, Cliff's $\delta = 0.30$, $p < 10^{-300}$; 70% exceed the clean median and 15% lie beyond the clean 95th percentile). This is direct, confound-controlled evidence for a genuine in-vitro-divergence component in the off-manifold compartment, over and above the coverage gap implied by the protocol gradient.

Together these diagnostics establish that the dominant off-target compartment is a reproducible biological mixture: a *reference-coverage gap* (non-cortical and later neural states absent from a first-trimester reference) superimposed on a pervasive *in-vitro stress* program (hypoxia, unfolded-protein response, glycolysis) that intensifies with maturation.

2.3 Lineage cannot be disentangled from in-vitro state in a shared latent (a characterized negative)

If the in-vitro state were a low-dimensional nuisance, one could subtract it and place off-target cells on a clean lineage axis. We tested this directly with contrastiveVI, which factorizes a background (shared) latent z_{lin} from a salient latent z_{iv} [28], training the background on primary fetal cells and the salient on organoid cells, so that z_{lin} should encode dish-stripped lineage. We evaluated disentanglement with a held-out adversary: a classifier that attempts to recover organoid-versus-primary origin from z_{lin} (balanced accuracy 0.5 is ideal; < 0.6 would indicate success) (Fig. 4a).

Disentanglement failed, and did so robustly. The baseline model ($\dim z_{\text{iv}} = 10$) yielded adversary balanced accuracy 0.998; tripling salient capacity ($\dim z_{\text{iv}} = 30$) gave 0.999; and adding a source batch covariate to the decoder—the textbook remedy for a nuisance axis confounded with a condition—gave 0.998. Training converged normally in every case (Fig. 4b), so this is not an optimization failure. The mechanism is structural and, given how different the two sources are, largely *expected*: the contribution here is to characterize precisely *why* a shared latent cannot work, not to present the failure as a surprise. When organoid and fetal cells occupy near-disjoint regions of expression space, source identity is almost definitionally decodable and there is no overlap region in which a shared lineage axis is even well defined—the background *encoder* $q(z_{\text{lin}} \,|\, x)$ observes and encodes the organoid-versus-primary difference directly, and conditioning the *decoder* on a batch label cannot undo what the encoder has already written into z_{lin} (Fig. 4c). Two scope limits on the interpretation follow, and we are explicit about them. First, the adversary establishes only that this difference is decodable, not what it is made of: organoid and primary datasets differ in biological in-vitro state *and* in laboratory, dissociation, platform, and ambient-RNA profile, every one of which is transcriptome-pervasive and would on its own render source decodable. The experiment therefore supports a claim we can fully defend—shared-latent contrastive methods cannot produce a source-invariant lineage axis in this regime—whereas the specifically *in-vitro* character of the dominant axis is established not by the adversary but by the stress diagnostics of the previous section (Fig. 3). Second, the donor-keyed integration used for mapping does not correct the far deeper organoid-versus-primary divide, so that divide is precisely what the background latent inherits. We accordingly state the negative result methodologically and read its biology from Aim 2's signatures. The substantive claim is therefore not that a domain gap exists—that is assumed—but a sharper, identifiability-flavoured one: shared-latent calibration *without anchoring* is feasible only where the two domains share an overlap region in which a common lineage axis is even defined, and the fetal-versus-organoid regime lies past that boundary. Naming this boundary condition—rather than restating the bare fact that source is decodable—is the contribution, and it identifies encoder-side or distributional methods as the only routes that could cross it.

2.4 Off-target populations occupy distinct, biologically coherent geometry

Having quantified the largely neural off-manifold compartment in Aim 2, we turn to the part of the off-target landscape where the reference is complete and the populations are large enough to support geometry: the *non-neural* contaminant lineages. This division of labor is deliberate. The neural off-manifold cells are the larger compartment, but, as Aim 2 showed, a first-trimester reference cannot yet resolve them into clean lineage structure (it conflates genuine in-vitro divergence with its own coverage gaps); the non-neural contaminants—skeletal and cardiac muscle, fibroblast, and adrenal-cortical cells—instead map to a *complete* fetal-body reference and number in the thousands, making them the compartment on which the geometry method can be put to a fair test. The disentanglement failure does not invalidate the geometry of these populations relative to one another. Aim 3 compares off-target *organoid* populations against *each other*; if all such cells share the same organoid-versus-primary offset c in z_{lin}, that offset is a constant translation that cancels exactly in pairwise sliced-Wasserstein distances ($W(\mathcal{A}{+}c, \mathcal{B}{+}c) = W(\mathcal{A}, \mathcal{B})$) [29], so the shared contamination does not distort relative structure.

This cancellation rests on one explicit assumption that we state plainly because the entire Aim 3 result depends on it: the source offset c must be approximately the *same* vector across the populations compared. That constancy is not guaranteed, and it is not the constancy Gate B verified—Gate B established offset-consistency among *on-target neural* cell types (cosine 0.855), whereas the cancellation is applied to *off-target non-neural* populations, a different set of cells. The stress biology cuts against strict constancy: if the unfolded-protein-response component scales with secretory demand, the offset should be larger for secretory lineages (e.g., adrenal-cortical) than for muscle, so part of a recovered "lineage" distance could in principle be differential-stress distance. Rather than rely on the constant-offset assumption, we tested it directly—measuring the per-lineage dish vector (organoid minus primary) for each of the four lineages and asking whether they coincide (Methods)—and report the outcome with the distances below, where it bounds how the geometry may be read.

We computed pairwise lineage distances among the true off-target populations, controlling for sample size by a rarefaction-derived minimum ($N_{\text{min}} = 2000$) and a matched-N self-distance noise floor; a pair is called only if its cross-distance confidence interval clears the floor (Fig. 5). Of the 17 off-target populations, four *non-neural contaminant* lineages are well powered (cortical-adrenal 49,338; skeletal-muscle 30,629; fibroblast 6,349; cardiac-muscle 2,534). All six pairwise comparisons are real differences, each clearing the noise floor by 14 to 105× (W from 0.32 to 0.99 against floors of 0.007–0.026); the remaining 130 small-N comparisons are correctly returned as underpowered.

The direct test, however, shows the constant-offset assumption does *not* hold for these populations. The per-lineage dish vectors are mutually inconsistent (mean pairwise cosine 0.15 versus 0.855 for on-target neural types; only the two myogenic lineages align, at 0.55), and each pairwise offset difference is comparable in magnitude to the lineage separation itself (ratio ≈ 1). The individual distances are thus confounded by a lineage-dependent offset and cannot be read as clean lineage metrics—the inter-lineage *direction* survives only weakly (cosine 0.29 between the organoid and primary inter-lineage vectors). What *does* survive is the rank structure: across the six pairs the organoid and primary pairwise-distance matrices correlate $r = 0.90$, so the metric still orders the lineages as the primary reference does (the two myogenic populations closest at $W = 0.32$, fibroblast most distant at $W \approx 0.96$–0.99, the steroidogenic adrenal-cortical population between; Fig. 5a,d). We therefore present Aim 3 as a *capability with a reference-validated ordering, not a clean lineage geometry*: the instrument produces statistically real, floor-clearing distances whose *rank* recovers primary lineage structure, whereas recovering true per-pair lineage distances would require per-lineage offset removal or a matched-primary embedding. Ordering grossly distinct lineages is in any case a low-bar capability check rather than a biological finding; its value is to license turning the same construction on questions whose answer is not known in advance (below).

3 Discussion

NullState turns the off-target problem of neural organoids into a quantitative, atlas-anchored measurement. Three results have practical consequences. First, the off-manifold compartment is large (38.4%) but *legible*: it decomposes into a reference-coverage gap and an in-vitro stress program, each separately actionable. The reference-gap component argues for extending fetal references beyond the first trimester and beyond cortex before off-manifold scores are interpreted as "aberrant" for non-cortical protocols; the stress component, intensifying with age and metabolic load and mirroring Bhaduri et al. [6], provides a per-cell readout for evaluating stress-mitigation strategies. The ~240-fold protocol spread is tempting to read as a fidelity ranking, and we caution explicitly against that misuse: off-manifold burden measures distance from a *first-trimester, cortex-rich* reference, so a choroid-plexus protocol at 96.8% off-manifold is largely *reference-absent*—non-cortical and later than HDBCA covers—rather than 96.8% aberrant. The axis ranks "how far from early cortex," conflating genuine in-vitro defect with region and stage mismatch; it is a fidelity axis only for protocols that target the reference's own domain, or once the reference is broadened. Read otherwise, it would mis-rank faithfully regionalized organoids (choroid plexus, midbrain, thalamus) as the worst performers, which they are not.

Second, our negative disentanglement result is useful precisely because it is characterized, and we are deliberate about what it does and does not claim. The widespread intuition that a contrastive or batch-aware latent can "subtract the dish" presupposes that the organoid-versus-primary difference is a separable nuisance subspace. We show that in this regime it is not—it is a pervasive, encoder-visible shift—so shared-latent methods cannot produce a source-invariant lineage axis regardless of salient capacity or decoder batch conditioning. That methodological conclusion holds whether the shift is biological or technical in origin; the separate evidence that its dominant component is specifically the in-vitro stress state comes from the Aim 2 diagnostics, not from the adversary, and the two should not be conflated. Methods that intervene at the encoder (e.g., adversarial or invariance-penalized encoders) or that operate distributionally rather than per-cell are the appropriate next step.

Third, the quantity Aim 3 needs—the *relative* geometry of off-target populations—survives the disentanglement failure only in part. The cancellation argument requires a per-lineage-constant source offset, and a direct test shows that offset is *not* constant for these populations (mutual cosine 0.15, against 0.855 for on-target neural types), so the recovered distances are confounded. What is validated is their *rank order*, which recovers the primary lineage structure (distance-matrix $r = 0.90$). Aim 3 is therefore a working capability with a reference-anchored ordering—a comparison axis whose ranking can be trusted today—rather than a clean lineage geometry or a finished biological result; its metric awaits per-lineage offset removal or a matched-primary embedding.

Limitations follow directly from the analysis. Off-manifold status conflates reference gap with biological aberrance, so absolute off-target rates are reference-dependent and should be read comparatively; indeed the pervasive organoid-versus-primary source gap displaces even clean, HDBCA-validated cells from an independent atlas, so the *absolute* coverage-gap fraction is not directly measurable (Fig. 6). The protocol gradient and the clean-calibrated cross-atlas contrast are the available evidence, and they agree on a mixture rather than on either pole. The dish-vector and stress scores are signature-based proxies, partially corroborated by the maturation correlation and the independent-atlas divergence signal. The well-powered Aim 3 populations are non-neural off-target lineages; neural off-target geometry awaits a reference that closes the coverage gap. And, as shown in the Results, the cancellation that licenses Aim 3 assumes a per-lineage-constant source offset that a direct test finds does *not* hold (mutual cosine 0.15); the recovered distances are accordingly confounded, and only their rank order—validated against the primary reference ($r = 0.90$)—should be relied on.

Finally, and most consequentially, the disentanglement wall does not close the convergence question that motivated this work—it reroutes it. The same translation invariance that rescues Aim 3 is itself a way to ask whether cell types of different developmental origin converge, *without* ever constructing a source-invariant per-cell coordinate: to test whether, say, brain-derived and kidney-derived mesenchyme have arrived at the same state, one needs only their *relative* geometry in a space where the shared source offset cancels—the construction demonstrated here, with the proviso, made concrete by Aim 3, that the shared offset must actually be constant across the compared populations or be explicitly removed: a checkable condition, not a free lunch. A cross-germ-layer convergence study therefore runs through the cancellation logic and an offset-constancy check, not through repairing contrastive disentanglement; the negative result is not a detour away from that goal but the identification of which road does not reach it, together with a partly working vehicle—and a clear specification of the remaining repair—for the road that does. Framed as a statement about what is recoverable from which data—an identifiability boundary for calibration without anchoring—the characterized negative is a methods contribution in its own right.

4 Methods

Reference construction. Raw-count fetal data were retrieved from the CZ CELLxGENE Discover Census [15]: the full first-trimester HDBCA [10] as the neural core, and non-neural compartments of a fetal-body atlas [11] filtered at the Census level to the relevant ontology cell types (mesoderm, endoderm, neural-crest origins). Gene identifiers were harmonized to Ensembl; counts were verified as integer UMIs prior to training. A lineage *origin* label was assigned to each reference cell through a curated cell-type→origin map; only the generic "cell" type was permitted to remain unmapped.

Mapping by reference surgery. We trained scVI (400 epochs, donor batch key) and scANVI on the reference, then mapped the HNOCA query [9] by scArches architectural surgery, using the implementations in scvi-tools [12–14, 16]. Organoid raw UMI counts were taken from the CELLxGENE HNOCA copy. Surgery stability at atlas scale required (i) dropping non-UMI read-count cells (per-cell maximum count > 1000) and (ii) a learning-rate, gradient-clipping, and KL-warmup schedule; without these the negative-binomial encoder diverged to NaN.

Scoring and classification. For each query cell we computed a label-assignment entropy H over scANVI posteriors and an off-manifold reconstruction distance ρ in the latent space. Thresholds τ_H and τ_R were calibrated on a held-out 10% of reference cells (off-manifold nearest-neighbor distances fit excluding the held-out set to avoid self-neighbor deflation). The six-state taxonomy combines on/off-manifold status (ρ vs. τ_R), label confidence (H vs. τ_H), and the lineage origin of the assigned label.

Diagnostics. Gene-set scores (neural progenitor, neuron, glia, hypoxia [19], unfolded-protein response, glycolysis) were computed with `scanpy.tl.score_genes` on log-normalized expression. Maturation was assessed against organoid age and a detected-gene potency proxy [20]; protocol stratification used the atlas `assay_differentiation` annotation; cluster coherence used Leiden communities [25] on the scPoli embedding.

Disentanglement. contrastiveVI [28] was trained with primary fetal on-target cells as background and organoid on-target cells as target (100,000 each, 2000 highly variable genes, raw-count layer). Disentanglement was scored by a five-fold cross-validated logistic-regression adversary predicting organoid-versus-primary origin from the background latent z_{lin} (balanced accuracy). Interventions: salient dimension $10 \to 30$, and a source batch covariate registered through `setup_anndata`.

Second-reference test. To probe coverage gap versus divergence directly, we jointly integrated 30,000 ambiguous-novel and 15,000 clean-on-target neural organoid cells with 100,000 prenatal cells of an independent cortical atlas (Velmeshev *et al.* [27]; gestational "LMP-month" stages, drawn from the CELLxGENE Census) using scVI with the dataset as batch key (3000 shared highly variable genes, 30-dimensional latent). Each organoid cell's off-manifold score is its mean distance to the 15 nearest reference cells in the integrated latent. Because clean and ambiguous-novel cells share the same organoid-versus-reference displacement, the absolute on-manifold fraction is confounded; we therefore calibrate against the clean distribution and report the clean-versus-ambiguous-novel effect size (Mann–Whitney AUC, Cliff's δ) and the fraction of ambiguous-novel cells beyond clean percentiles.

Geometry. For each off-target population pair we estimated the sliced-Wasserstein distance [29] between background-latent embeddings, with a per-population minimum sample size from rarefaction ($N_{\min} = 2000$) and a matched-N self-distance noise floor (95th percentile). A pair was called a real difference when its bootstrap cross-distance interval exceeded the floor; multiplicity was controlled by Benjamini–Hochberg [30]. Sliced-Wasserstein pairwise distances are invariant to a translation shared by both populations, which is what makes the comparison valid despite the residual source offset in z_{lin}—*provided* that offset is the same across the populations compared.

Offset-constancy test. For each of the four well-powered off-target lineages we drew up to 2500 organoid cells (true-off-target, by predicted label) and the matched primary cells (same CELLxGENE ontology label), selected 2000 highly variable genes on the pooled raw counts, log-normalized (CP10k, $\log 1p$), and computed the per-lineage dish vector as the difference of organoid and primary centroids. We report the mutual cosine of these dish vectors (offset consistency), the cosine between the organoid and primary inter-lineage centroid vectors (geometry preservation), the ratio of the offset difference to the primary lineage separation, and the Pearson correlation of the organoid and primary pairwise-distance matrices across the six pairs (`scripts/dishvec_consistency.py`).

Compute and code. Models were trained on a single NVIDIA H100 GPU. All quantitative panels are generated directly from pipeline artifacts (`docs/manuscript/make_figures.py`); numerical values are exported to `numbers.json`.

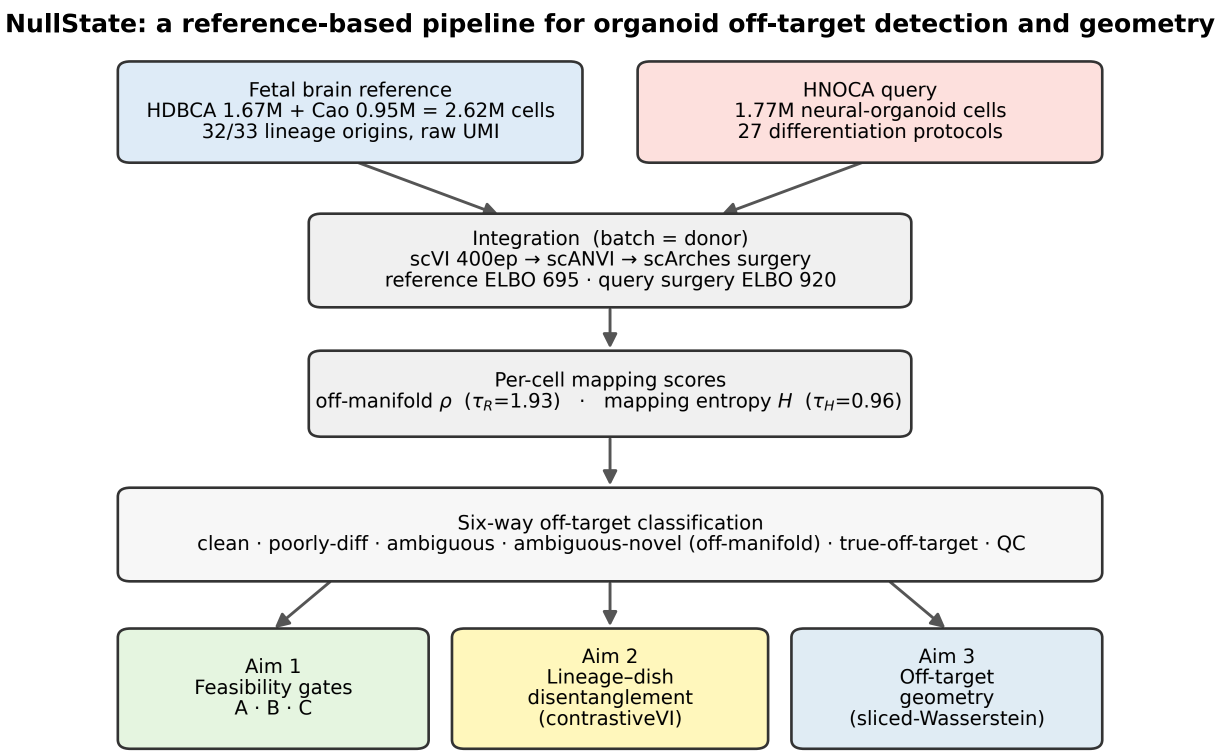

Figure 1. The NullState pipeline. A harmonized first-trimester fetal reference (HDBCA + fetal-body atlas; 2,615,898 cells, raw UMIs) and the HNOCA organoid query (1,767,346 cells, 27 protocols) are co-embedded by scVI/scANVI followed by scArches reference surgery. Each organoid cell receives two calibrated mapping scores—off-manifold distance ρ ($\tau_R = 1.93$) and label entropy H ($\tau_H = 0.96$)—that define a six-state off-target taxonomy, which feeds three analyses: feasibility gates (Aim 1), lineage–dish disentanglement (Aim 2), and off-target geometry (Aim 3).

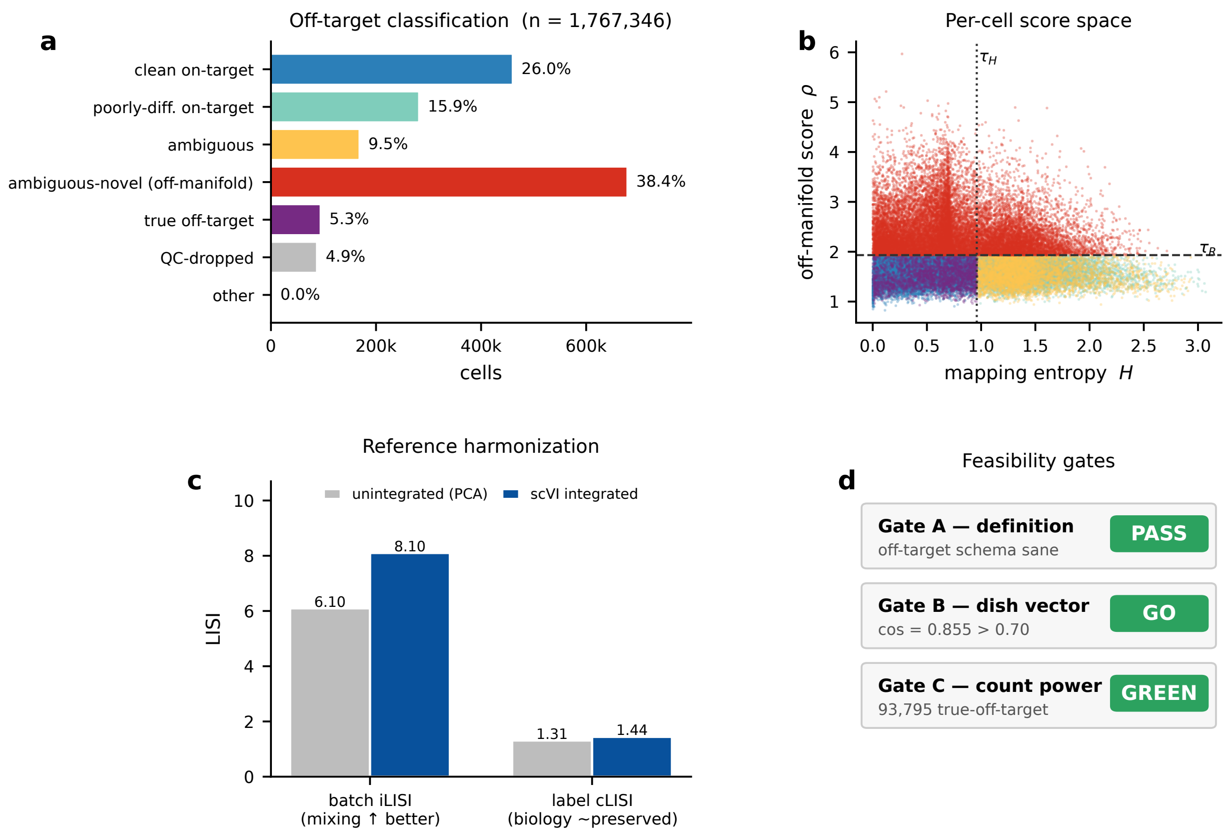

Figure 2. Harmonized mapping and the off-target landscape (Aim 1). (**a**) Six-state classification of all 1,767,346 organoid cells; the off-manifold *ambiguous-novel* state dominates (38.4%). (**b**) Per-cell score space: mapping entropy H versus off-manifold score ρ (subsample), coloured by class, with the calibrated thresholds τ_H (dotted) and τ_R (dashed). (**c**) Donor-batch integration improves mixing (iLISI $6.10 \to 8.10$) while preserving biology (cLISI $1.31 \to 1.44$). (**d**) The three feasibility gates pass: definition (A), dish-vector self-consistency (B, cosine 0.855), and off-target power (C, green).

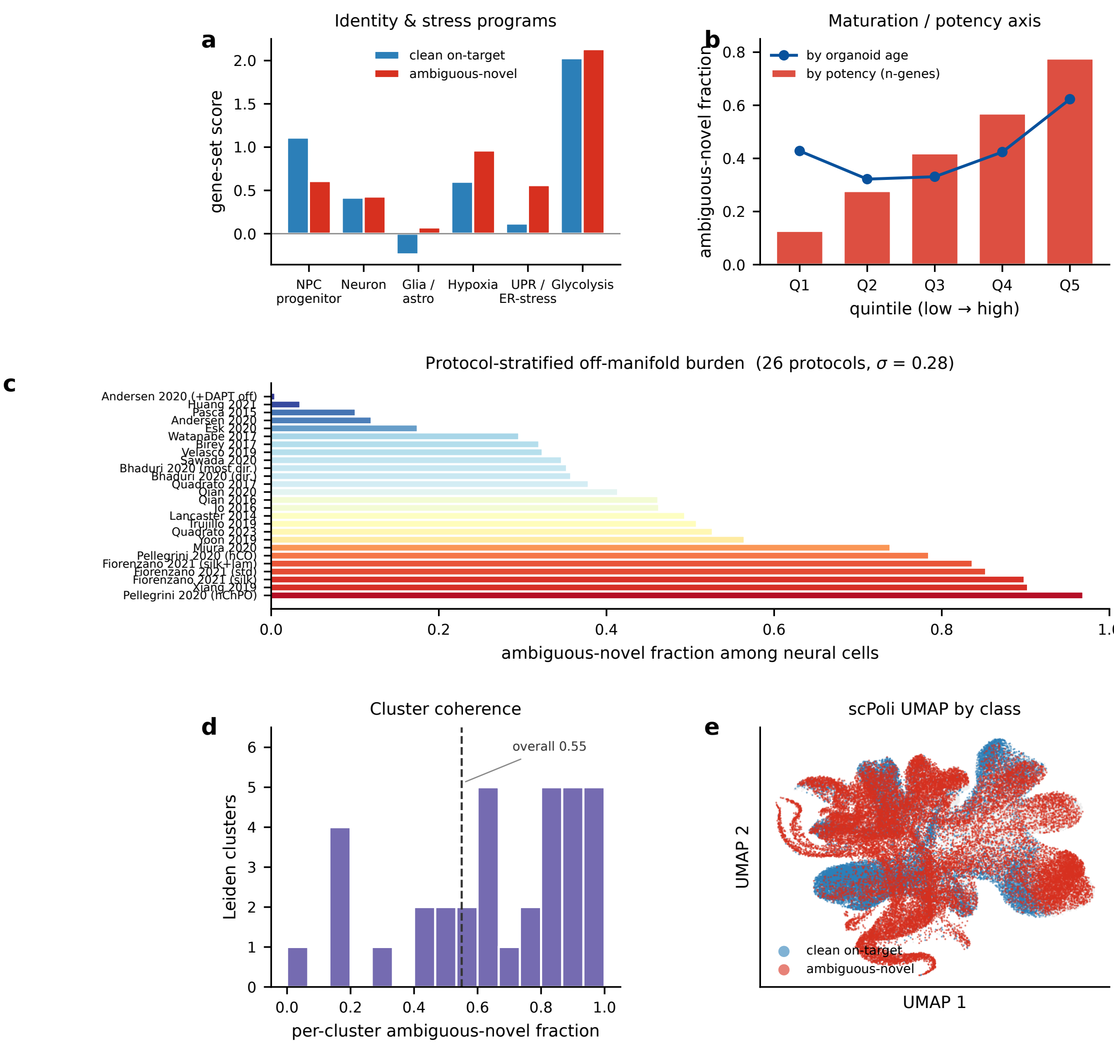

Figure 3. The off-manifold compartment is structured, not artifactual. (**a**) Ambiguous-novel neural cells lose neural-progenitor identity and gain hypoxia and unfolded-protein/ER-stress signatures relative to clean on-target neural cells. (**b**) Off-manifold fraction increases monotonically with a detected-gene potency proxy (bars) and with organoid age (line). (**c**) Protocol-stratified off-manifold burden spans 0.4%–96.8% across 26 protocols (colour = fraction); guided cortical protocols are lowest, choroid-plexus/midbrain/thalamic highest. (**d**) Per-cluster off-manifold fraction across 30 Leiden communities is bimodal (coherent populations, not scatter). (**e**) scPoli UMAP coloured by class: ambiguous-novel cells occupy contiguous territory.

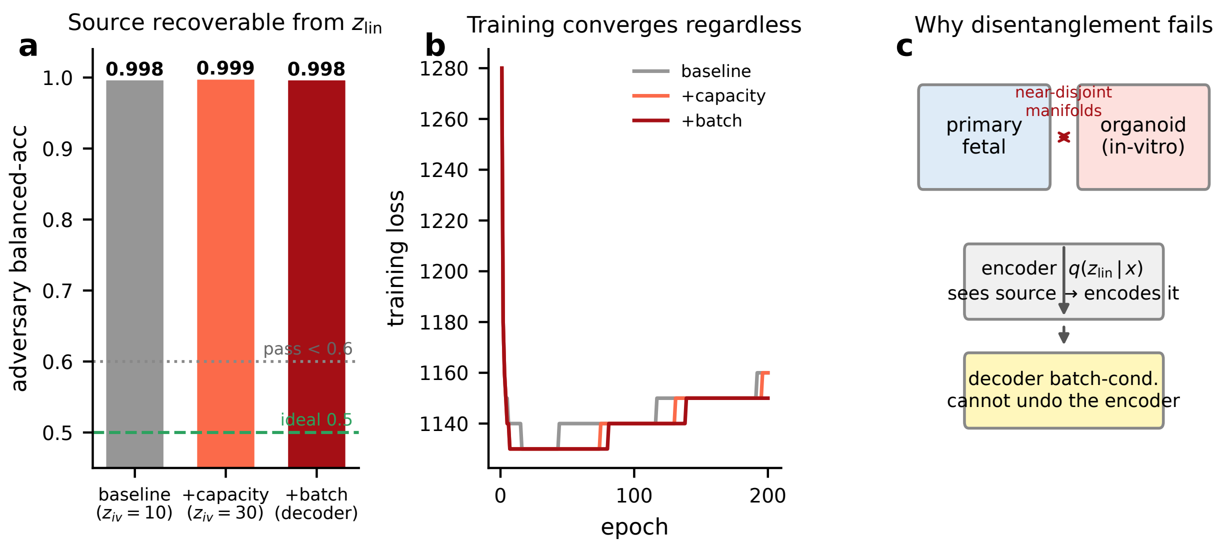

Figure 4. Lineage–dish disentanglement is not attainable (characterized negative; Aim 2). (**a**) An origin-decoding adversary recovers organoid-versus-primary identity from the "dish-stripped" latent z_{lin} at ≥ 0.998 balanced accuracy across all interventions (ideal 0.5; pass < 0.6). (**b**) Training converges normally in every run, so the failure is structural, not optimization-related. (**c**) Mechanism: near-disjoint primary/organoid manifolds make the background *encoder* encode source directly; decoder batch-conditioning cannot remove it.

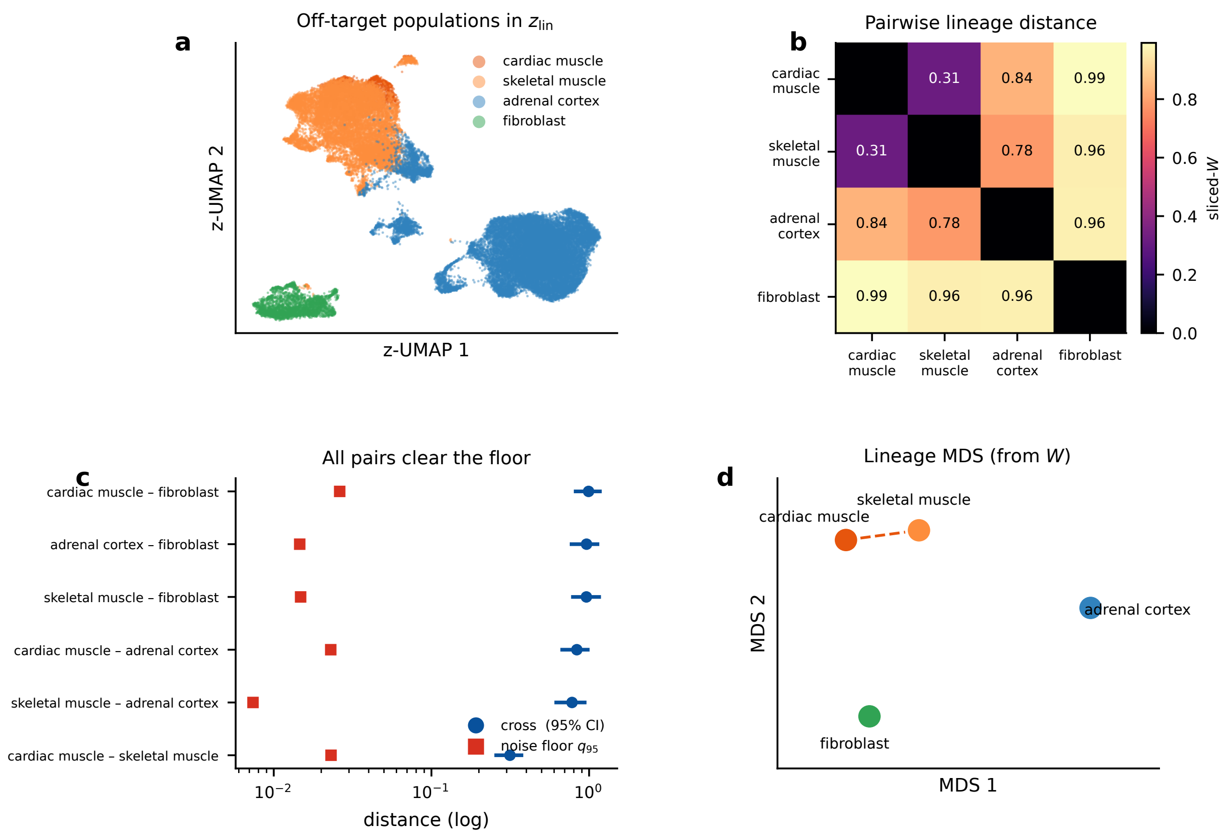

Figure 5. Off-target geometry: distances clear the noise floor and their rank recovers primary lineage order (Aim 3). (**a**) Background-latent (z_{lin}) embedding of the off-target populations; the four well-powered lineages separate. (**b**) Pairwise sliced-Wasserstein distance matrix. (**c**) Every well-powered pair clears its matched-N noise floor (q_{95}, red) by 14–105× (cross-distance with 95% CI, blue; log axis). (**d**) Multidimensional scaling of the distance matrix: the two myogenic populations (cardiac, skeletal muscle) are closest, fibroblast most distant. The rank order matches the primary lineage geometry (distance-matrix $r = 0.90$); the individual distances, however, are confounded by lineage- dependent source offsets (mutual cosine 0.15; see text), so the ordering is validated but the metric is not a clean lineage distance.

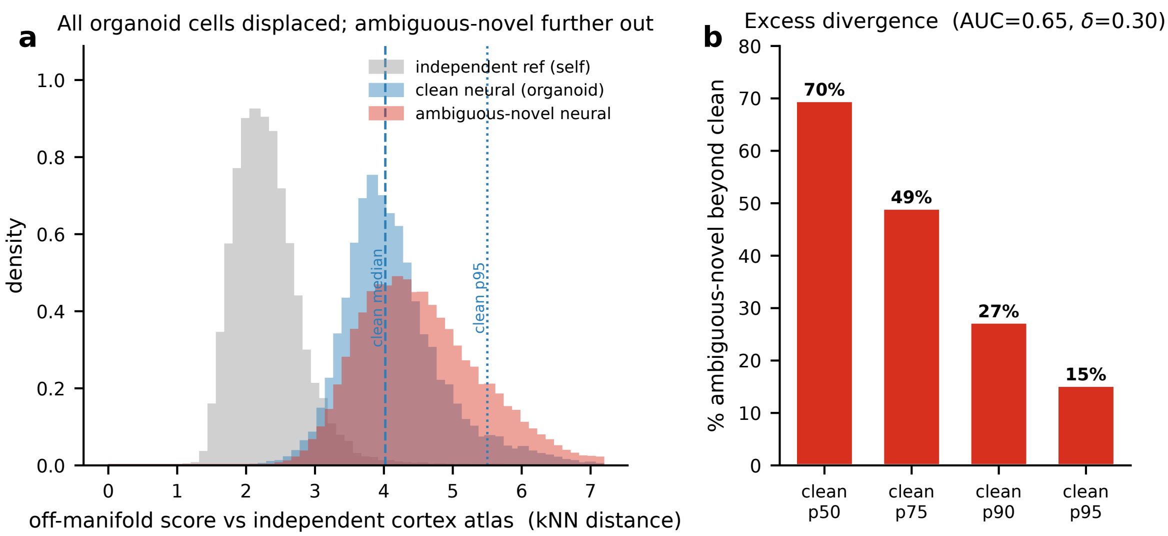

Figure 6. Direct cross-atlas test of coverage gap versus divergence. Ambiguous-novel and clean-on-target neural organoid cells were jointly integrated with an *independent* prenatal cortical atlas (Velmeshev *et al.* 2023 [27]). (**a**) Off-manifold score (kNN distance to the independent reference) for reference-self (grey), clean neural (blue), and ambiguous-novel neural (red). The organoid-versus-reference source gap displaces *all* organoid cells—including clean cells—far from the reference (clean/ambiguous medians 4.0/4.4 versus reference-self 2.2), so absolute mapped-on fractions are confounded; the clean distribution (dashed median, dotted 95th percentile) is the calibrator. (**b**) The clean-versus-ambiguous contrast cancels the shared displacement: 70% of ambiguous-novel cells exceed the clean median and 15% lie beyond the clean 95th percentile (overall AUC = 0.65, Cliff's $\delta = 0.30$), isolating a genuine in-vitro-divergence component.

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